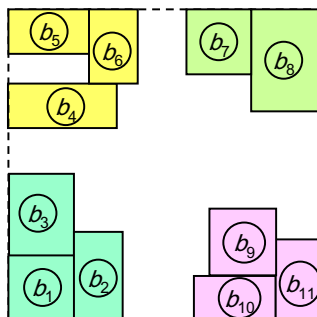
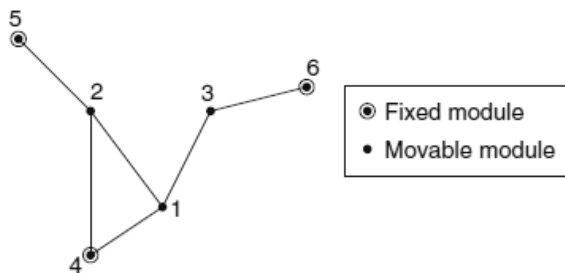


Final Exam

1. (10%) You are asked to place a cell i on a chip. The cell i connects to four other pads a, b, c , and d at the coordinates $(4, 9)$, $(12, 2)$, $(3, 18)$, and $(15, 6)$ with the weights 1, 2, 3, and 5, respectively. Find an appropriate position to place the cell i by using the force-directed method.
2. (20%) Consider the placement of the modules shown below.
 - (a) Give the MP-tree for the module placement.
 - (b) Discuss how you can utilize only two contours (could be a horizontal or a vertical contour) to pack the modules into the four corners. Show your packing step by step. What is the time complexity of your packing scheme?

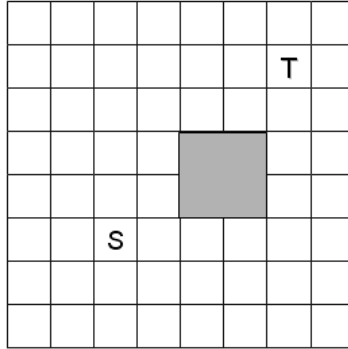


3. (10%) Consider the example shown below. Assume all nets have a weight of 1. For the three fixed modules, module 4 is at $(6, 2)$, module 5 is at $(0, 9)$, and module 6 is at $(8, 6)$. Determine the locations of the movable modules such that the total weighted quadratic wirelength is minimized.

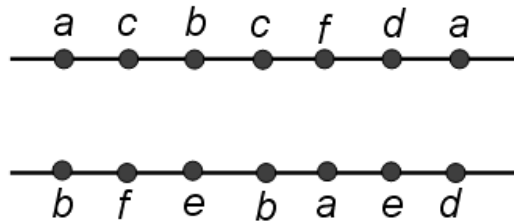


4. (20%) Give five points $a = (1, 20)$, $b = (6, 18)$, $c = (12, 9)$, $d = (8, 16)$, and $e = (8, 2)$ in the plane.
 - (a) Apply the method discussed in class to construct a spanning graph for the five points. **Show your work!**
 - (b) Apply the point-edge connection method to construct a Steiner tree. What is the resulting tree cost? **Show your work!**

5. (10%) Find the path from S to T by the A*-search routing for the instance shown below by by defining $g(x)$ to be the label from the source S to the current node of x and $h(x)$ to be the Manhattan distance between x and the target T , where the cost function is given by $f(x) = g(x) + h(x)$. Break the tie by picking the grid cell on the right of the current position. **(Please show you work and labels step by step.)**



6. (20%) Given the instance of the channel routing problem shown below.
- Determine the maximal sets and give a tight lower bound on the channel width.
 - Draw the HCG and VCG.
 - Can the constrained left-edge algorithm apply to this channel routing instance? Route the instance if this figure gives a feasible routing instance; explain why the algorithm does not apply to this instance, otherwise.
 - Can the dogleg channel router apply to this channel routing instance? Route the instance if this figure gives a feasible routing instance by alternating the routing sequence, top, bottom, top, bottom, ...; explain why the algorithm does not apply to this instance, otherwise.



7. (10%) Given a signal wire (a, b) of length $l_{ab} = 5000\mu m$ and width $2\mu m$. Let the unit wire resistance and capacitance be $\alpha = 0.1\Omega/\mu m$ and $\beta = 0.2fF/\mu m^2$, and unit buffer resistance and capacitance be $\alpha' = 100\Omega/\text{unit size}$ and $\beta' = 20fF/\text{unit size}$, respectively. Assume that driver resistance seen at the source a is $R_d = 100\Omega$, and the capacitive load $C_L = 20fF$ seen at the sink b . Apply the π model for wires and the switch-resistor model for buffers to compute the Elmore delay for the following instances. If the buffer size is twice of the unit-sized one, where should we insert the buffer to minimize the delay for the buffered wire? What is the resulting delay? Show your work.